

Minsung Kim

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Education

University of Toronto - Computer Science Specialist

2025 - 2029

GPA 3.7/4 | *University of Toronto International Scholar Award*

Skills

Languages: C/C++, C#, Python, Java, TypeScript, JavaScript, SQL, HTML5/CSS, PostgreSQL, MySQL, C/C++, C#, Bash, R

Tools/Frameworks: React, RESTful APIs, PostgreSQL, Git, GitHub Actions, Linux, Postman, AWS, Tailwind CSS, Next.js, jQuery

Experience

ResNet Technical Assistant

May 2026 – Present

University of Toronto

- Troubleshoot Wi-Fi, Ethernet, and eduroam connectivity issues across Chestnut, Oak House, and Graduate House, resolving network access problems for residential users
- Assist residents with wired and wireless device connectivity, including Ethernet cables, USB-A/USB-C adapters, HDMI cables, and gaming consoles requiring MAC address registration
- Diagnose eduroam authentication issues on Android devices by configuring certificates and selecting appropriate CA settings or providing guest-network alternatives when necessary
- Monitor residential network infrastructure through regular access-point and connectivity checks, while restoring disconnected Ethernet and university phone lines through troubleshooting and power cycling

Project

UofT Student Life Tracker

July 2026 – August 2026

- Developed a student dashboard integrating UofT resources using React, Express, PostgreSQL, and Material UI to centralize the navigation between separate platforms
- Built a RESTful API calendar and deadline system that allows users to create events, tasks, and deadlines, organizing academic planning into a single website
- Designed the interface using Material UI and implemented user authentication with bcrypt for password encrypting

Hack the Student Life

March 2026

AWS Services & University of Toronto

- Developed ClusterGrade AI with a team of three, an AI grading platform that extracts text from scanned handwritten assignments using AWS, Python Scikit-Learn, and Sentence Transformers to automate analysis of student submissions
- Built an OCR-to-clustering pipeline that converts scanned assignments into semantic embeddings and applies K-Means to group submissions by recurring logic errors, with the local prototype achieving a 0.72 silhouette score across 10 test submissions.
- Integrated Amazon Bedrock with instructor-provided rubrics to generate rubric-aligned deductions and feedback for clustered submissions, enabling one TA review to be distributed across entire groups of students rather than manually rewriting identical feedback
- Designed a human-in-the-loop grading workflow that requires TA approval before applying AI-generated feedback to clustered submissions, targeting a 40–60% reduction in grading time for large introductory courses.

Hack Canada

March 2026

SPUR Innovation

- Developed WatchLine as a team of three, a real-time AI transit safety system using YOLOv8-Pose, Python, and open CV to track human pose and classify falls, aggression, crouching, lying down, and erratic movement from live camera streams
 - Built an asynchronous FastAPI and WebSocket backend that processes live video, maintains per-person motion histories, automatically records critical incidents as video clips, and broadcasts detection events to connected clients in real time.
 - Integrated Google Gemini 2.0 Flash to convert video evidence of detected incidents into English dispatch summaries, while Supabase/PostgreSQL stores incident data and provides access to recorded events.
- Developed a React, Vite, and Zustand monitoring dashboard that renders MJPEG camera streams, displays real-time safety alerts, and provides incident clip playback for rapid event review.